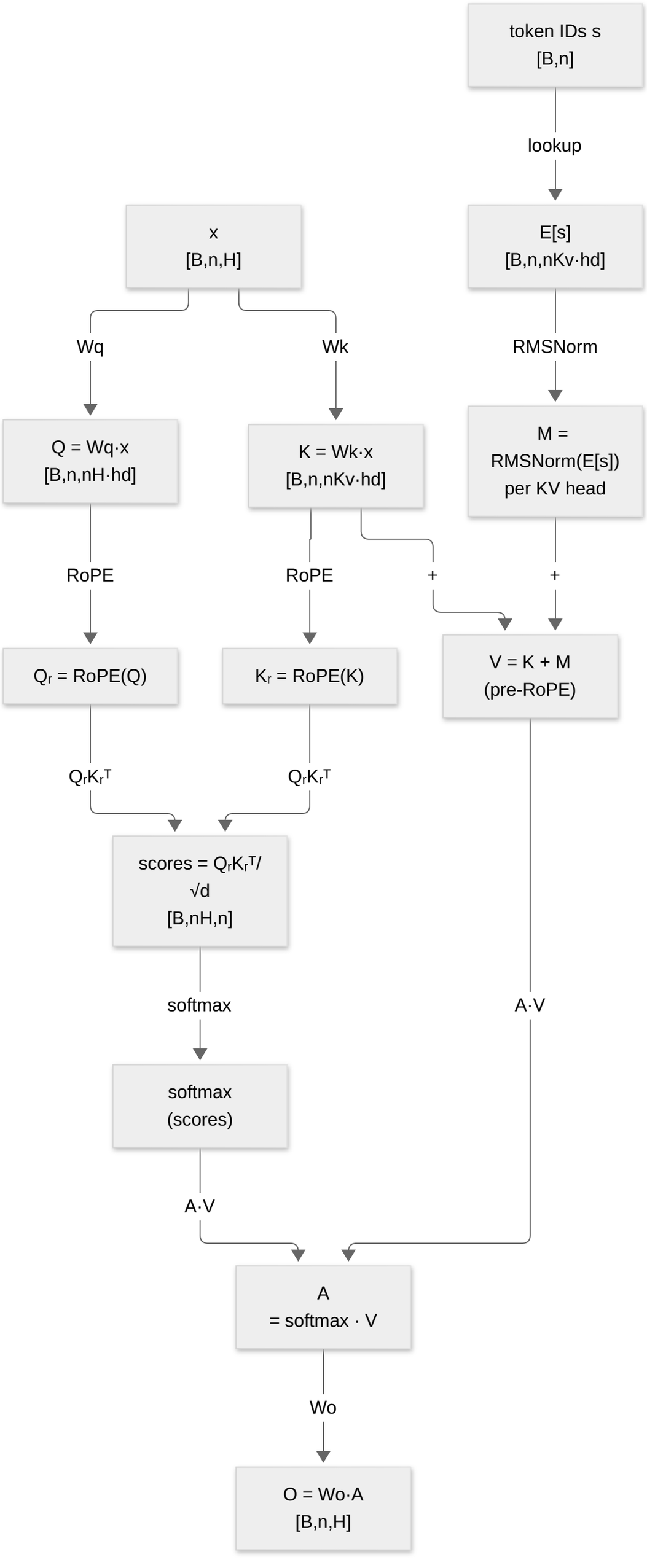

token IDs s
[B,n]
lookup
x
[B,n,H]
E[s]
[B,n,nKv·hd]
Wq
Wk
RMSNorm
Q = Wq·x
[B,n,nH·hd]
K = Wk·x
[B,n,nKv·hd]
M =
RMSNorm(E[s])
per KV head
RoPE
RoPE
+
+
Qr = RoPE(Q)
Kr = RoPE(K)
V = K + M
(pre-RoPE)
QrKrT
QrKrT
scores = QrKrT/
√d
[B,nH,n]
softmax
A·V
softmax
(scores)
A·V
A
= softmax · V
Wo
O = Wo·A
[B,n,H]